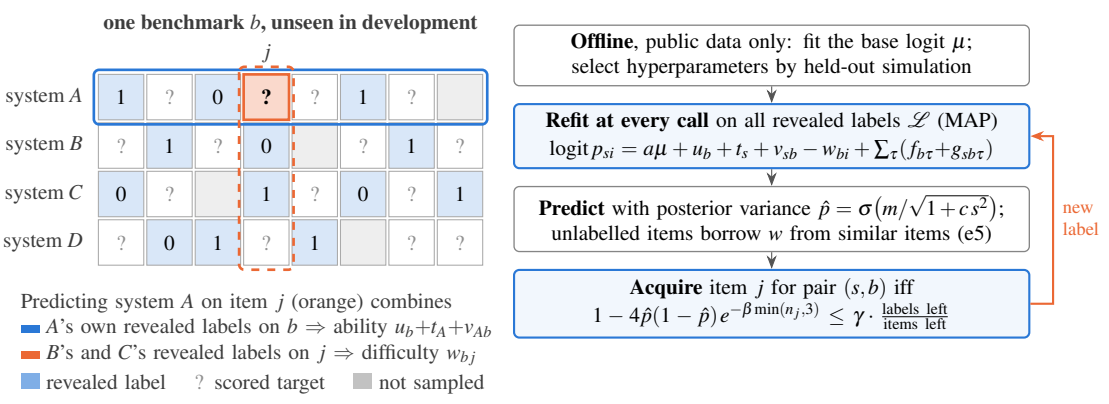

one benchmark b, unseen in development
j
system A
system B
system C
system D
1 ? 0 ? ? 1 ?
? 1 ? 0 ? 1 ?
0 ? 1 ? 0 ? 1
? 0 1 ? 1 ? ?
Predicting system A on item j (orange) combines
A's own revealed labels on b ⇒ ability $u_b+t_A+v_{Ab}$
B's and C's revealed labels on j ⇒ difficulty w_{bj}
revealed label
? scored target
not sampled
Offline, public data only: fit the base logit μ; select hyperparameters by held-out simulation
Refit at every call on all revealed labels $\mathscr{L}$ (MAP)
$\operatorname{logit} p_{si} = a\mu + u_b + t_s + v_{sb} - w_{bi} + \sum_\tau (f_{b\tau} + g_{sb\tau})$
Predict with posterior variance $\hat{p} = \sigma\left(m/\sqrt{1+cs^2}\right)$; unlabelled items borrow w from similar items (e5)
Acquire item j for pair (s,b) iff
$1 - 4\hat{p}(1-\hat{p})\,e^{-\beta \min(n_j,3)} \le \gamma \cdot \frac{\text{labels left}}{\text{items left}}$
new label